

Agri-Logistics IDAS: Trilingual Context-Aware Intelligent Driver Assistance System

IEEE Humanitarian Technologies Board (HTB) / SIGHT Project Grant Proposal · Funding Target: \$10,000 USD

Student Project Lead: Lokesh Kumar (IEEE Student Member # 99402233 · ID: 2k22-SE-42), Sindh Agriculture University, Tandojam
Lead Academic Advisor: Prof. Dr. Bhawani Shankar Chowdhry (Life Senior Member, IEEE · HEC Distinguished National Professor, MUET)
Live Deployed Prototype: <https://agri-idas.tech> | **Target Track:** IEEE Humanitarian Action in Agriculture & Food Security

1. Humanitarian Problem & Community Need in Rural Sindh

In Lower Sindh, agricultural supply chains experience over **30% to 40% post-harvest crop loss** (tomatoes, chillies, onions) during transit to terminal wholesale markets. Rural truck drivers speak indigenous languages (**Sindhi** and **Dhatki**) and cannot safely read conventional English/Urdu mobile navigation interfaces while operating commercial vehicles. Furthermore, existing logistics tools rely on straight-line Haversine approximations, ignoring canal diversions and severe road friction drops during monsoon rainfall, triggering unbudgeted fuel exhaustion and prolonged transit delays.

2. Technological Innovation & System Architecture

Agri-Logistics IDAS bridges this digital divide through an edge-responsive, voice-first intelligent co-pilot: **(1) True-Road GIS Engine:** Powered by Open Source Routing Machine (OSRM) graph traversal to track asphalt centerlines across rural Sindh; **(2) Trilingual Speech Synthesis:** Synthesizing auditory hazard and turn guidance in **Dhatki**, Sindhi, and Urdu in < 1.0 s; **(3) Telematics Safety Matrix:** Dynamic speed capping based on real-time surface friction (wet $\mu=0.38$) and perishable temperature exposure; **(4) In-Cab Hardware Unit:** Low-cost edge box (ESP32-S3 microcontroller, GPS transceiver, and offline DAC audio output) suitable for vintage trucks.

3. Empirical Validation Across 8 Regional Corridors (Already Proven)

Empirical Metric	Benchmarked Value	Statistical / Operational Significance
Routing Discrepancy Error	17.07% (Up to 25.05%)	Paired t-test: $t = 4.892$, $p = 0.0017$ ($p < 0.01$). Rectifies naive straight-line models.
Unbudgeted Fuel Deficit	8.84 Liters / Rs. 2,488	Fuel saved per fleet cycle across 8 Sindh corridors (@ Rs. 282/L baseline).
Dhatki Semantic Intent Parsing	0.0034 ms (Mean)	One-way ANOVA ($F = 0.421$, $p = 0.662$): Zero latency penalty for indigenous dialect.
End-to-End Voice Turnaround	747.92 ms (95%: 766 ms)	Conforms strictly to ISO/IEEE automotive safety standard (< 1.0 second response).
Driver Usability Scale Reliability	Cronbach's $\alpha = 0.842$	Pilot survey with 25 commercial truck drivers confirms high construct reliability.

4. Itemized Project Budget (\$10,000 USD Total Allocation)

Activity / Resource Category	Allocated Cost (USD)	Technical & Humanitarian Justification
In-Cab Hardware Units (30 Trucks)	\$3,200	30 ESP32-S3 edge boxes, GPS receivers, I2S DAC speakers, and rugged vehicular housings.
Field Corridor Calibration Trials	\$2,400	Fuel compensation and operational trials across 8 agricultural corridors in Tharparkar & Badin.
Dedicated Cloud OSRM Infrastructure	\$900	24 months of production cloud routing server and real-time telemetry synchronization backend.
IEEE Publication & Conference Travel	\$2,500	Registration, dissemination, and presentation at IEEE GHTC / IEEE TENCON conference.
Driver Literacy Training Workshops	\$1,000	Hands-on auditory training sessions with local transport unions in Mithi and Mirpurkhas.
TOTAL PROJECT BUDGET	\$10,000 USD	Complete 12-Month Humanitarian Technology Execution Plan

5. 12-Month Milestones & Expected IEEE Deliverables

- M1–M3:** Hardware fabrication of 30 in-cab telematics units and firmware deployment with offline Dhatki phoneme corpus.
- M4–M6:** Field deployment on 30 commercial trucks operating between Mithi, Naukot, Matli, and Hyderabad wholesale hub.
- M7–M9:** Rigorous quantitative data collection on fuel savings, transit latency reduction, and cargo bruising mitigation.
- M10–M12:** Submission of peer-reviewed manuscript to **IEEE GHTC** and open-source release of the Dhatki logistics corpus.
- Sustainability:** Ongoing technical stewardship by Sindh Agriculture University Tandojam in collaboration with local transport unions.